THE TWO STAR-COMPOSITES A STAR (B_N) AND (A_N) STAR B ARE HOMOTOPIC THROUGH A_{I-1} STAR B_{N-I}

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof uses the 1_a shorthand introduced in the proof of lemma-godement and the rewriting identities for d^n_j and s^n_j are displayed in-span before use; both target composites and their analogues on G' circ X circ F' are handled symmetrically.

Editorial note. The label lemma-godement-two-maps names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The Godement functorial setup. Same chapter, \label{example-godement-functorial}. Line 6612: the categories, Y, d, s and the composable functors F, G with X the simplicial endofunctor.
- The relations satisfied by a simplicial homotopy. Same chapter, \label{lemma-relations-homotopy}. Lines 6627-6632: conditions (1)-(5) that the family h_{n,i} must satisfy; each is verified below.
- The alternative homotopy formalism. Same chapter, Remark \label{remark-homotopy-better}. Line 6631: cross-reference for the homotopy relations used.
- The interleaving calculus for d^n_j and s^n_j. Same chapter, \label{lemma-godement} (its proof introduces the 1_a shorthand). Lines 6640-6652 and 6684-6694: d^n_j and s^n_j rewritten by sliding the middle d resp. s across the 1-factors.

Source reference. Original reference 0G5P: sources/stacks-simplicial/pinned-original-source.tex, source lines 6563–6563. Prerequisite pointer invoked at proof line(s) 6614; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0G5N: sources/stacks-simplicial/pinned-original-source.tex, source lines 6443–6443. Prerequisite pointer invoked at proof line(s) 6651; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 019L: sources/stacks-simplicial/pinned-original-source.tex, source lines 4704–4704. Prerequisite pointer invoked at proof line(s) 6633; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 019M: sources/stacks-simplicial/pinned-original-source.tex, source lines 4737–4737. Prerequisite pointer invoked at proof line(s) 6634; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$, Y, d, s, F, G be as in Example 0G5P. Let $F' : \mathcal{A} \to \mathcal{C}$ and $G' : \mathcal{C} \to \mathcal{B}$ be two functors. Let $(a_n) : G \circ X \to G' \circ X$ be a morphism of simplicial objects compatible via augmentations with $a : G \to G'$. Let $(b_n) : X \circ F \to X \circ F'$ be a morphism of simplicial objects compatible via augmentations with $b : F \to F'$. Then the two maps*

$$a \star (b_n), (a_n) \star b : G \circ X \circ F \to G' \circ X \circ F'$$

are homotopic.

Proof. To show the morphisms are homotopic we construct morphisms

$$h_{n,i} : G \circ X_n \circ F \to G' \circ X_n \circ F'$$

for $n \geq 0$ and $0 \leq i \leq n+1$ satisfying the relations described in Lemma 019L. See also Remark 019M. To satisfy condition (1) of Lemma 019L we are forced to set $h_{n,0} = a \star b_n$ and $h_{n,n+1} = a_n \star b$. Thus a logical choice is

$$h_{n,i} = a_{i-1} \star b_{n-i}$$

for $1 \leq i \leq n$. Setting $a = a_{-1}$ and $b = b_{-1}$ we see the displayed formular holds for $0 \leq i \leq n+1$.

Recall that

$$d^n_j = 1_G \star 1_j \star d \star 1_{n-j} \star 1_F$$

on $G \circ X \circ F$ where we use the notation $1_a = 1_{Y \circ \ldots \circ Y}$ introduced in the proof of Lemma 0G5N. We are going to use below that we can rewrite this as

$$\begin{aligned} d^n_j &= d^j_j \star 1_{n-j} = d^{j+1}_j \star 1_{n-j} = \ldots = d^{n-1}_j \star 1_1 \\ &= 1_j \star d^{n-j}_0 = 1_{j-1} \star d^{n-j+1}_1 = \ldots = 1_1 \star d^{n-1}_{j-1} \end{aligned}$$

Of course we have the analogous formulae for d^n_j on $G' \circ X \circ F'$.

We check condition (2) of Lemma 019L. Let $i > j$. We have to show

$$d^n_j \circ (a_{i-1} \star b_{n-i}) = (a_{i-2} \star b_{n-i}) \circ d^n_j$$

Since $i - 1 \geq j$ we can use one of the possible descriptions of d^n_j to rewrite the left hand side as

$$(d^{i-1}_j \star 1_{n-i+1}) \circ (a_{i-1} \star b_{n-i}) = (d^{i-1}_j \circ a_{i-1}) \star b_{n-i} = (a_{i-2} \circ d^{i-1}_j) \star b_{n-i}$$

Similarly the right hand side becomes

$$(a_{i-2} \star b_{n-i}) \circ (d_j^{i-1} \star 1_{n-i+1}) = (a_{i-2} \circ d_j^{i-1}) \star b_{n-i}$$

Thus we obtain the same result and (2) is checked.

We check condition (3) of Lemma 019L. Let $i \leq j$. We have to show

$$d_j^n \circ (a_{i-1} \star b_{n-i}) = (a_{i-1} \star b_{n-1-i}) \circ d_j^n$$

Since $j \geq i$ we may rewrite the left hand side as

$$(1_i \star d_{j-i}^{n-i}) \circ (a_{i-1} \star b_{n-i}) = a_{i-1} \star (b_{n-1-i} \circ d_{j-i}^{n-i})$$

A similar manipulation shows this agrees with the right hand side.

Recall that

$$s_j^n = 1_G \star 1_j \star s \star 1_{n-j} \star 1_F$$

on $G \circ X \circ F$. We are going to use below that we can rewrite this as

$$\begin{aligned} s_j^n &= s_j^j \star 1_{n-j} = s_j^{j+1} \star 1_{n-j-1} = \ldots = s_j^{n-1} \star 1_1 \\ &= 1_j \star s_0^{n-j} = 1_{j-1} \star s_1^{n-j+1} = \ldots = 1_1 \star s_{j-1}^{n-1} \end{aligned}$$

Of course we have the analogous formulae for s_j^n on $G' \circ X \circ F'$.

We check condition (4) of Lemma 019L. Let $i > j$. We have to show

$$s_j^n \circ (a_{i-1} \star b_{n-i}) = (a_i \star b_{n-i}) \circ s_j^n$$

Since $i - 1 \geq j$ we can rewrite the left hand side as

$$(s_j^{i-1} \star 1_{n-i+1}) \circ (a_{i-1} \star b_{n-i}) = (s_j^{i-1} \circ a_{i-1}) \star b_{n-i} = (a_i \circ s_j^{i-1}) \star b_{n-i}$$

Similarly the right hand side becomes

$$(a_i \star b_{n-i}) \circ (s_j^{i-1} \star 1_{n-i+1}) = (a_i \circ s_j^{i-1}) \star b_{n-i}$$

as desired.

We check condition (5) of Lemma 019L. Let $i \leq j$. We have to show

$$s_j^n \circ (a_{i-1} \star b_{n-i}) = (a_{i-1} \star b_{n+1-i}) \circ s_j^n$$

This equality holds because both sides evaluate to $a_{i-1} \star (s_{j-i}^{n-i} \circ b_{n-i}) = a_{i-1} \star (b_{n+1-i} \circ s_{j-i}^{n-i})$ by exactly the same arguments as above. □